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Min Woo Park, Ph.D.

SHORT BIO. During my Ph.D., I focused on *causal decision-making* (NeurIPS-25, UAI-26a, ICLR-26) and *causal effect identification* (NeurIPS-24, UAI-26c). I investigated how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations.

In particular, I developed methods for robust decision-making under uncertainty (NeurIPS-25), domain shifts (UAI-26a), and even counterfactual actions (ICLR-26).

EDUCATION

Graduate School of Data Science, Seoul National University — Seoul, Republic of Korea
Ph.D. in Data Science — Sep. 2022 – Feb. 2027
Advisor: Prof. Sanghack Lee.
Thesis: *Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals*.
Doctoral Qualifying Exam waived due to outstanding academic performance.

Department of Mathematics and Statistics, Sejong University — Seoul, Republic of Korea
B.S. in Applied Statistics and Mathematics (Double major), *Cum Laude* — Mar. 2016 – Aug. 2022
Advisors: Prof. Ji Eun Lee and Jin Hee Yoon. — (2.5 years of military service)

RESEARCH EXPERIENCE

Samsung Research | AI Research Engineer — Starting in Mar. 2027
Furiosa AI | AI Research Engineer Internship — Jun. 2026 – Sep. 2026
Causality Lab, Seoul National University | Ph.D. Researcher — Jan. 2023 – Feb. 2027
Functional Analysis Lab, Sejong University | Undergraduate Researcher — Jul. 2021 – Dec. 2022
ISDS Lab, Sejong University | Undergraduate Researcher — Jan. 2021 – Dec. 2022

PUBLICATIONS

* for joint authorship and † for corresponding author.

◇ Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals.
Min Woo Park and Sanghack Lee†.
Causality Lab, Seoul National University, Technical Report, 2026.

◇ On Transportability for Structural Causal Bandits.
Min Woo Park and Sanghack Lee†.
Uncertainty in Artificial Intelligence (UAI), 2026.

◇ Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery.
Min Woo Park*, Taehui Yun*, Youngin Jang, Younseok Yeom, Jonghwan Kim, Jiyeon Kang, Songseong Kim, Hyemin Jung, Sangmin Lee, Jongseong Jang†, and Sanghack Lee†.
Uncertainty in Artificial Intelligence (UAI), 2026.

◇ Canonical Domain Reduction for Partial Counterfactual Identification.
Yesong Choe, Yeahoon Kwon*, **Min Woo Park***, and Sanghack Lee†.
Uncertainty in Artificial Intelligence (UAI), 2026.

◇ Counterfactual Structural Causal Bandits.
Min Woo Park and Sanghack Lee†.
International Conference on Learning Representations (ICLR), 2026.

◇ Structural Causal Bandits under Markov Equivalence.
Min Woo Park, Andy Arditi, Elias Bareinboim†, Sanghack Lee†.
Neural Information Processing Systems (NeurIPS), 2025.
CausalAI Laboratory, Columbia University, Technical Report (R-122).

◇ Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference.
Yonghan Jung†, **Min Woo Park**, and Sanghack Lee†.
Neural Information Processing Systems (NeurIPS), 2024.

◇ Computation of the iterated Aluthge, Duggal, and Mean transforms.
Min Woo Park* and Ji Eun Lee*†.
Filomat, 2023.

◇ Convergence of the iterated mean transforms of a 2×2 matrix.
Ji Eun Lee*† and **Min Woo Park***.
under review.

AWARDS AND HONORS

NRF Ph.D. fellowship in science and engineering, 2026 | National Research Foundation of Korea
▷ *Active Learning-Based Action Space Optimization for Causal Multi-Armed Bandits under Incomplete Information*. Sep. 2026 – Aug. 2027

UAI 2026 scholarship Aug. 2026

Graduate Student Instructor (GSI) scholarship Spring 2024, Spring 2025, Spring 2026

BK21 FOUR scholarship Fall 2024, Fall 2025

First place merit scholarship | Sejong University Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University
▷ *Mitigating Spurious Regression in Time-Series Data via Granger Causality*. Nov. 2021

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University
▷ *The Significance of Euler's Number and Its Applications*. Nov. 2020

PROJECTS

Towards More Scalable Score-based Causal Discovery | LG AI Research Sep. 2025 – Aug. 2026
▷ Leading a project on scalable score-based causal discovery algorithms.
- Developed a novel operator enabling score-based causal discovery algorithms to escape local optima.
- Investigating parallelization strategies for score-based causal discovery algorithms.

▷ Resulting paper accepted at *Uncertainty in Artificial Intelligence* (UAI-26b).
- Software available at `https://github.com/LGAI-Research/breaking-bad`.

▷ Korean patent application filed.

Hyundai NGV Data Boot-Campus | Hyundai Motor Company
▷ **2025.** Developed high-voltage battery performance prediction technology. Jul. 2025 – Sep. 2025
▷ **2024.** Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometry–based automatic curvature analysis. Jul. 2024 – Sep. 2024

Optimal Mixed Policies Completeness | Oxford University Aug. 2023 – Feb. 2024
▷ Studied characterization of conditions under which an intervention effect of a variable is positive.

TEACHING ASSISTANT EXPERIENCE

Data Science Project Fall 2026
▷ Practical experience in end-to-end project execution.

Causal Inference for Data Science Spring 2024, Fall 2024, Spring 2025, Spring 2026
▷ Topics included Bayesian networks and graphical causal inference.

Machine Learning and Deep Learning for Data Science II Fall 2023
▷ Covered kernel methods, Gaussian processes, and related topics in machine learning.

Basic Mathematics for Artificial Intelligence (Undergraduate) Spring 2022
▷ Covered linear algebra, basic statistics, and probability theory.

ACADEMIC SERVICES, PROFESSIONAL TALKS AND POSTERS

Academic Services

Journal Reviewer:

Journal of the American Statistical Association (JASA)

The Journal of Artificial Intelligence (AIJ)

Conference Reviewer: **2027** ICLR, **2026** NeurIPS, UAI, ICLR, ICML, **2025** ICML

Reviewed AI/ML lecture video content from LG AI Research for LG Aimers
Summer 2025, Winter 2025, Summer 2026

Invited Talks

Major Exploration for Sejong University Students | Seoul, Republic of Korea — Nov. 2026

SNU GSDS student research seminar | Seoul, Republic of Korea — Dec. 2024, Apr. 2025

Poster Sessions

UAI 2026 poster session | Amsterdam, Netherlands — Aug. 2026

ICLR 2026 poster session | Rio de Janeiro, Brazil — Apr. 2026

ExploreCSR poster session | Seoul, Republic of Korea — Dec. 2025

NeurIPS 2025 poster session | San Diego, USA — Dec. 2025

JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea — Nov. 2025

SNU GSDS workshop poster session | Jeju, Republic of Korea — Jul. 2025

NeurIPS 2024 poster session | Vancouver, Canada — Dec. 2024

REFERENCES

Sanghack Lee | Graduate School of Data Science, Seoul National University — sanghack@snu.ac.kr
Associate professor

Ji Eun Lee | Department of Mathematics and Statistics, Sejong University — jieunlee7@sejong.ac.kr
Associate professor

Jin Hee Yoon | Dept. of AI and Information Technology, Sejong University — jin9135@sejong.ac.kr
Associate professor

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